



National University of Computer and Emerging Sciences, Lahore



Build Hub - Build Your Dreams

Ali Hamza 19L-2204 BS(CS)

Hafiz Abdul Mannan 19L-1080 BS(CS)

Hassan Khalid 19L-1181 BS(CS)

Supervisor:

Hafiz Muhammad Usama Hassan Alvi

Final Year Project

May 1, 2024

Anti-Plagiarism Declaration

This is to declare that the above publication was produced under the:

Title: Build Hub - Build Your Dreams

is the sole contribution of the author(s), and no part hereof has been reproduced as it is the basis (cut and paste) that can be considered Plagiarism. All referenced parts have been used to argue the idea and cited properly. I/We will be responsible and liable for any consequence if a violation of this declaration is determined.

Date: October 8th, 2023

Name: Ali Hamza

Signature: alihamza

Name: Abdul Manan

Signature: Manana

Name: Hassan Khalid

Signature: hassan

Author's Declaration

This states Authors' declaration that the work presented in the report is their own, and has not been submitted/presented previously to any other institution or organization.

Abstract

Finding the ideal firm for the construction industry has proven to be challenging. Traditionally, individuals contact builders one at a time to choose the best one based on their past experiences. In this situation, there is a possibility that people may try to exploit someone by posing as a phony builder. Build Hub can assist individuals in connecting with construction companies quickly in the age of expanding internet use and convenience shopping. We are automating the process by creating a platform where builders can register their companies on and client can contact the builders of the specific area. The project includes features such as cost estimation, profile management, seamless communication, reviewing system and an AI driven chatbot. The project aims to improve and simplify user experience, and promote effective platform interactions.

Executive Summary

Even in the era of technological growth, many people still find it difficult to choose the best construction company. Individual contacts with builders were generally made, with selections based on prior encounters. Such a strategy not only requires time, but also raises the possibility of deception by dishonest constructors. Build Hub, a cutting-edge web-based platform created to transform the relationship between people and construction enterprises, takes on this task. Clients may easily locate and contact builders based on particular requirements and areas by allowing builders to register their businesses on the site.

Build Hub's two ground-breaking machine learning-driven features set it distinct from conventional approaches. A sophisticated matching mechanism for clients and builders is the first. In order to provide a match that guarantees effectiveness and satisfaction, this system intelligently assesses the project specifications established by the customer against the portfolios of registered builders. The second function streamlines the typically time-consuming ID verification process. Build Hub guarantees that client IDs and land papers are validated effortlessly by utilizing state-of-the-art Optical Character Recognition (OCR) and image processing techniques, encouraging confidence and security among its customers.

Build Hub's goals go beyond only bringing together clients and contractors. The platform aims to be a complete answer that can meet all of the demands of its user base. Builders are given a set of tools that enable them to maintain their profiles, supervise projects, communicate with clients, and even securely handle payments. Customers gain from user-friendly features that expedite project posting, builder communication, contract signing, and service assessments, on the other side. The platform also incorporates administrative features, allowing for effective user verification and administration.

The preliminary research emphasizes the importance of Build Hub and its potential for success. The desire for digital solutions is clear given that there are over 13,000 smartphone applications in the construction business. These applications have revolutionized several aspects of the construction industry, including design changes, site monitoring, worker productivity, and weather forecasts. More significantly, they have filled the knowledge gap that frequently results in expensive reworks and misunderstandings. In this context, Build Hub seeks to further streamline interactions by providing a platform where strong documentation and verification procedures are combined with efficient communication.

In conclusion, Build Hub is the next development in the construction sector's digital progression. It claims to transform how people and businesses interact by fusing cutting-edge technologies with user-centric design, assuring efficiency, transparency, and trust.

Table of Contents

List of Figures **ix**

List of Tables **xi**

1 Introduction **1**

1.1 Purpose of this Document 1

1.2 Intended Audience 1

1.3 Definitions, Acronyms, and Abbreviations 1

1.4 Conclusion 2

2 Project Vision **3**

2.1 Problem Domain Overview 3

2.2 Problem Statement 3

2.3 Problem Elaboration 3

2.4 Goals and Objectives 3

2.5 Project Scope 4

2.6 Sustainable Development Goal (SDG) 4

2.7 Constraints 4

2.8 Business Opportunity 5

2.9 Stakeholders Description/ User Characteristics 5

2.9.1 Stakeholders Summary 5

2.9.2 Key High-Level Goals and Problems of Stakeholders 5

2.10 Conclusion 5

3 Literature Review / Related Work **6**

3.1 Definitions, Acronyms, and Abbreviations 6

3.2 Detailed Literature Review 6

3.2.1 Digital Platforms in Construction Management 6

3.2.2 Construction Tech: The Digital Transformation 7

3.2.3 Machine Learning in Construction Planning 7
3.2.4 OCR for Document Verification in Real Estate 8
3.2.5 Internet's Role in Construction Management 9
3.2.6 Smart Recommendations in Real Estate Platforms 9
3.2.7 Construction Project Management Software: A Review 10
3.2.8 Sustainable Construction Practices: A Review of Recent Developments 11
3.2.9 IoT Applications in Construction: A Comprehensive Review 11
3.2.10 BIM in Construction: State-of-the-Art and Future Directions 12
3.3 Literature Review Summary Table 13
3.4 Conclusion 14

4 Software Requirement Specifications 15
4.1 List of Features 15
4.2 Functional Requirements 15
4.3 Quality Attributes 16
4.4 Non-Functional Requirements 16
4.5 Assumptions 17
4.6 Use Cases 17
4.7 Hardware and Software Requirements 21
4.7.1 Hardware Requirements 21
4.7.2 Software Requirements 21
4.8 Graphical User Interface 21
4.9 Database Design 22
4.9.1 ER Diagram 23
4.9.2 Data Dictionary 23
4.10 Risk Analysis 24
4.11 Conclusion 24

5 High-Level and Low-Level Design 25
5.1 System Overview 25
5.1.1 Builders Profile 25
5.1.2 Proposal Submission and Builder Verification 25
5.1.3 Project Bidding by Builders 25
5.1.4 Project Progress Tracking 25
5.1.5 Quality Assurance and Documentation 25

5.1.6 Cost Estimation of the Project . . . 26
5.2 Design Considerations . . . 26
5.2.1 Assumptions and Dependencies . . . 26
5.2.2 General Constraints . . . 26
5.2.3 Goals and Guidelines . . . 28
5.2.4 Development Methods . . . 28
5.3 System Architecture . . . 28
5.3.1 Project Matching and Posting Module . . . 29
5.3.2 Proposal Submission and Builder Verification Module . . . 29
5.3.3 Project Bidding by Builders Module . . . 30
5.3.4 Project Progress Tracking Module . . . 30
5.3.5 Quality Assurance and Documentation Module . . . 30
5.3.6 Cost Estimation of the Project Module . . . 30
5.3.7 Server . . . 30
5.4 Architectural Strategies . . . 30
5.4.1 Programming Language . . . 31
5.4.2 Product Enhancement and Extensibility . . . 31
5.4.3 Concurrency and Synchronization . . . 31
5.5 Domain Model/Class Diagram . . . 32
5.6 Sequence Diagrams . . . 32
5.7 Policies and Tactics . . . 36
5.7.1 Tools to be Used . . . 36
5.7.2 Coding Guidelines and Conventions . . . 36
5.7.3 User Interface . . . 36
5.7.4 Extensibility . . . 36
5.7.5 Policies for System Testing . . . 36
5.7.6 Software Maintenance . . . 36
5.7.7 Algorithms . . . 37
5.8 Conclusion . . . 37

6 Implementation and Test Cases 38

6.1 Implementation . . . 38
6.1.1 Login/SignUp . . . 38
6.1.2 Prototype description . . . 38
6.1.3 Backend . . . 39

6.2 Test Case Design and Description 39
6.2.1 Client Side Test Cases 39
6.2.2 Builder Side Test Cases 46
6.3 Test Metrics 49

7 User Manual **50**
7.1 Client 50
7.1.1 Getting Started 50
7.1.2 Create New Project 50
7.1.3 Check Project Status 50
7.1.4 Chat with Others 50
7.1.5 Bidding 50
7.1.6 Use ChatBot 51
7.1.7 Give Ratings to Builders 51
7.2 For Builder 51
7.2.1 Getting Started 51
7.2.2 Give Response to Bids 51

8 Conclusion and work Completed **52**

List of Figures

4.1 **LogIn Page** 21
4.2 **SignUp Page** 22
4.3 **Home Page** 22
4.4 **ER Diagram** 23

5.1 **System Architecture diagram** 29
5.2 **Class Diagram** 32
5.3 **Login Process** 32
5.4 **Sign-Up Procedure** 33
5.5 **View Project Details** 33
5.6 **Delete Project Operation** 34
5.7 **Edit Project Data** 34
5.8 **Add Project Workflow** 35
5.9 **Check Credentials** 35

List of Tables

3.1 **Summary of Literature Review for Development Projects** 13

4.1 **User Login** 17
4.2 **User Registration** 18
4.3 **View Lands** 18
4.4 **Add Land Record** 18
4.5 **Edit Land Record** 19
4.6 **Delete Land Record** 19
4.7 **Sign Rental Agreement** 19
4.8 **Verify Land Record** 20
4.9 **Search Lands** 20
4.10 **Leave Feedback** 20
4.11 **Data Dictionary for User Table** 23
4.12 **Data Dictionary for Project Table** 24
4.13 **Data Dictionary for Agreement Table** 24
4.14 **Data Dictionary for Land Record Table** 24

6.1 **Login Test Case** 39
6.2 **Sign Up Test Case** 40
6.3 **Update Profile Test Case** 40
6.4 **View Profile Test Case** 41
6.5 **Forget Password Test Case** 41
6.6 **Search Builder Test Case** 42
6.7 **View Builder Test Case** 42
6.8 **Add Project Test Case** 43
6.9 **Get Projects Test Case** 43
6.10 **View Project Test Case** 44
6.11 **Get Messages Test Case** 44
6.12 **View Message Test Case** 45

6.13 **Add Bidding Test Case** 45
6.14 **Get Clients Test Case** 46
6.15 **View Client Test Case** 46
6.16 **Get Notifications Test Case** 47
6.17 **View Notification Test Case** 47
6.18 **Check Project Progress Test Case** 48
6.19 **Search Client Test Case** 48
6.20 **Logout Test Case** 49
6.21 **Functionality Metrics** 49

Chapter 1 Introduction

Finding the best business and guaranteeing the integrity of its dealings are issues for the construction industry, which has historically relied on direct connections between builders and clients. The ”Build Hub” approach, which aims to change these interactions by utilizing technology, particularly machine learning and OCR verification, is further described in this document.

1.1 Purpose of this Document

Specify the purpose of this Project. This document's goal is to give a thorough description of our Final Year Project (FYP), which is focused on developing the ”Build Hub” platform, which links clients and builders in the construction sector. This platform provides an efficient route for communication while also introducing cutting-edge capabilities like OCR for document verification and intelligent suggestions powered by machine learning. Can we create a technologically sophisticated platform that promotes sincere, effective, and smooth interactions between builders and clients? This is the key research issue covered in this article. The project's techniques, design, implementation plans, testing phases, outcomes, constraints, and potential future improvements are explained in this report.

1.2 Intended Audience

The academic panel reviewing this Final Year Project, construction industry stakeholders interested in technological advancements, potential investors looking at novel construction tech solutions, and future researchers hoping to build upon this work are the main audiences for this report.

1.3 Definitions, Acronyms, and Abbreviations

Build Hub: The primary platform discussed in this project connecting builders and clients.
ML: Machine Learning.
OCR: Optical Character Recognition.
FYP: Final Year Project.
UX: User Experience
UI: User Interface.
API: Application Programming Interface.

1.4 Conclusion

In Conclusion, ”Build Hub” is ready to transform the construction industry by fusing a user-friendly front-end interface with a reliable back-end system. In addition to streamlining interactions between builders and clients, the platform makes use of cutting-edge technology like machine learning and optical character recognition (OCR) to improve user experience and guarantee data validity. The team is dedicated to creating a platform that strikes a balance between functionality and security, ensuring that every stakeholder benefits from its services. This will be accomplished by thorough study and by taking inspiration from existing systems.

Chapter 2 Project Vision

Sectors that have historically been resistant to technological change now have unprecedented prospects because to the fast changing digital world. ”Build Hub” aims to take advantage of this potential to transform the building sector. We go more deeply into the project’s aims, larger objectives, and the issue area in this chapter.

2.1 Problem Domain Overview

The construction sector, which is well-known for making considerable economic contributions, continues to rely significantly on conventional operating procedures. These antiquated procedures frequently result in waste and are vulnerable to fraud. ”Build Hub” envisions a platform that connects builders and clients that is simplified, secure, and supported by machine learning and OCR technologies with powerful capabilities.

2.2 Problem Statement

The lack of a centralized, technologically driven ecosystem in the construction sector results in a fragmented builder-client contact process that is rife with inefficiencies, opaque, and open to fraud.

2.3 Problem Elaboration

The contacts between builders and clients are dominated by manual processes, which makes it difficult to verify the legitimacy of builders and may result in frauds. Customers end up making poor judgments since they are frequently unaware of their options. On the other hand, reliable builders lose out on opportunities because of poor exposure. These difficulties are made worse by the lack of intelligent recommendation tools and the human document verification procedure.

2.4 Goals and Objectives

Create the ”Build Hub” platform, a angular-based platform that provides a comprehensive remedy for the construction sector. Adapt machine learning algorithms to match intelligently builders and clients.

- Tailor an all-in-one solution for Builders, Customers, and Admin in construction, meeting all their needs.

- Empower Builders with tools for portfolio management, project assignments, and secure payments.
- Offer user-friendly features for project management and communication, enhancing Customer confidence.
- Implement advanced tech, like machine learning and OCR verification, for efficient and secure processes.

2.5 Project Scope

”Build Hub” is intended to serve as the go-to resource for all parties involved in the building sector:

- A platform built on the angular stack, SQL Server, dart, and .net that ensures scalability, flexibility, and a smooth user experience.
- Systems that provide intelligent recommendations using machine learning.
- OCR-based automated document verification minimizes mistakes and user involvement. ensuring top-notch data security procedures and fostering user confidence.
- Although the initiative offers game-changing solutions, it’s important to stress that its goal is to improve the first builder-client interactions rather than to replace actual building labor.

2.6 Sustainable Development Goal (SDG)

Industry, Innovation, and Infrastructure is demonstrated via ”Build Hub”. The platform enhances the value of resilient infrastructure, advances sustainable industrialisation, and fosters innovation by digitalizing a crucial industry like building.

2.7 Constraints

The seamless integration of machine learning and OCR with the angular stack presents technical obstacles. overcoming obstacles from builders who still operate traditionally. ensuring dependable data quality and accuracy for document verification. navigating the intricate regulatory environment that surrounds the construction sector.

2.8 Business Opportunity

The wave of digital change offers ”Build Hub” a tremendous commercial potential. As clients and builders look more and more for digital solutions, the platform may establish itself as the standard for the sector by providing unmatched features and advantages. The angular stack-powered platform's modular structure also enables potential future extensions and integrations.

2.9 Stakeholders Description/ User Characteristics

Builders: Professionals keen on expanding their clientele and embracing digital transformations. Clients: Individuals or entities eager for a streamlined, secure, and transparent builder selection process. Administrators: Custodians of platform integrity, ensuring a flawless user experience

2.9.1 Stakeholders Summary

”Build Hub” is designed to meet the specific demands of all stakeholders, from modern builders to traditionalists, knowledgeable clients to first-timers. The website claims to be inclusive and to serve a wide range of users.

2.9.2 Key High-Level Goals and Problems of Stakeholders

Efficiency, openness, and security are priorities for stakeholders. ”Build Hub” aims to close the gaps now present in the building sector by giving users a platform that is not only technologically cutting edge but also simple to use.

2.10 Conclusion

Build Hub's tech-driven platform, which combines OCR and machine learning, has the potential to completely transform the construction industry. The project provides builders, clients, and administrators with a holistic solution that prioritizes efficiency and security. Congruent with the Industry, Innovation, and Infrastructure Sustainable Development Goal, ”Build Hub” seeks to act as a stimulant for constructive transformation within the building sector. Although there are obstacles to overcome, the project sees a profitable commercial potential in the industry's digital transformation, and its modular platform design allows for future flexibility. The inclusive approach envisions a future when technology enriches conventional building methods and sets new benchmarks for quality, all while catering to a broad user base.

Chapter 3 Literature Review / Related Work

The development of ”Build Hub” is contingent upon an understanding and assessment of significant advancements in the domain of digital construction management. This chapter examines the body of literature, shedding light on pertinent research and projects that form the basis of our work.

3.1 Definitions, Acronyms, and Abbreviations

- angular stands for SQL Server, dart, and .net stack for developing web applications.
- OCR stands for optical character recognition, a tool for identifying text included inside digital images.
- SDG: Sustainable Development Goal.

3.2 Detailed Literature Review

3.2.1 Digital Platforms in Construction Management

3.2.1.1 Summary of the research item

The revolutionary potential of digital platforms in construction management is explored in the work by A. Smith and B. Johnson [1]. It emphasizes how these platforms may increase transparency, efficiency, and collaboration while reducing common problems like misunderstandings, holdups, and overruns.

3.2.1.2 Critical analysis of the research item

The following list outlines the strength and weakness of this platform that we found:

- Thorough understanding of the use of digital platforms in building.
- Instances of effective applications in the real world.
- Restricted to a certain technical environment.
- Lacks a user-centered viewpoint.

3.2.1.3 Relationship to the proposed research work

The guiding ideas presented in this study serve as the inspiration for our initiative, ”Build Hub” [1]. Even while the paper stresses the strength of digital platforms, ”Build Hub” goes above and beyond by using machine learning and OCR technologies, offering a more complete solution.

3.2.2 Construction Tech: The Digital Transformation

3.2.2.1 Summary of the research item

The digital revolution of the construction industry is examined in L. Anderson's article [2]. It describes how technological developments are changing conventional construction methods and examines the possible advantages, such as enhanced effectiveness and cost savings.

3.2.2.2 Critical analysis of the research item

The following list outlines the strength and weakness of this platform that we found:

- Provides a thorough overview of the digital transition in the building industry.
- Discusses success stories and real-world applications.
- Highlights the advantages of possible cost savings.
- Lacks in-depth knowledge of certain technology and techniques powering the transition.
- There is insufficient treatment of implementation problems and potential obstacles.

3.2.2.3 Relationship to the proposed research work

The research conducted by Anderson fits within the larger framework of "Build Hub" [2]. It recognizes the revolutionary potential of technology in building, which is consistent with the goal of our project. "Build Hub" will enlarge on certain technology and offer workable answers to problems.

3.2.3 Machine Learning in Construction Planning

3.2.3.1 Summary of the research item

The study of R. Adams and S. Patel focuses on using machine learning in building planning [3]. It investigates how scheduling and resource allocation for building projects might be optimized using machine learning methods.

3.2.3.2 Critical analysis of the research item

The following list outlines the strength and weakness of this platform that we found:

- A thorough investigation of machine learning's potential for planning building projects.
- Gives real-world examples of how to use machine learning to improve project scheduling.
- Draws attention to the possibility of cost and time savings.

- The difficulties and restrictions of applying ML in building have received less attention.
- Lacks a more comprehensive understanding of technology integration than planning.

3.2.3.3 Relationship to the proposed research work

The ”Build Hub” initiative can immediately benefit from Adams and Patel’s study, particularly when it comes to streamlining the building process [3]. In line with the results of this study, ”Build Hub” proposes to implement machine learning for predictive analytics and resource optimization.

3.2.4 OCR for Document Verification in Real Estate

3.2.4.1 Summary of the research item

Optical Character Recognition (OCR) technology is the main topic of D. Murphy’s study on document verification in the real estate industry [4]. It demonstrates how OCR may improve accuracy and productivity when dealing with real estate paperwork.

3.2.4.2 Critical analysis of the research item

The following list outlines the strength and weakness of this platform that we found:

- Gives an explicit and useful use case for OCR technology in the real estate industry.
- Demonstrates the potential to reduce document verification mistakes.
- Focuses on the significance of accurate data in real estate deals.
- Limited discussion of implementation-related difficulties with OCR, such as processing documents with handwriting.
- Lacks a comparison of available OCR options.

3.2.4.3 Relationship to the proposed research work

A significant part of ”Build Hub” is directly informed by Murphy’s study [4]. OCR technology is used by ”Build Hub” to verify documents, confirming the veracity and correctness of IDs and land papers. This is consistent with the research’s focus on enhancing real estate document management.

3.2.5 Internet's Role in Construction Management

3.2.5.1 Summary of the research item

E. White's study explores how the internet affects construction management [5]. It examines how advancements in communication and internet platforms are changing the building sector.

3.2.5.2 Critical analysis of the research item

The following list outlines the strength and weakness of this platform that we found:

- Gives information on the rising significance of internet platforms in the building industry.
- Demonstrates the possibilities for enhanced cooperation and communication.
- Acknowledges the importance of exchanging real-time data.
- Limited discussion of issues with data security and privacy related to internet platforms.
- Places greater emphasis on the benefits than any potential drawbacks.

3.2.5.3 Relationship to the proposed research work

White's study fits in with the "Build Hub" project's larger framework [5]. It highlights the value of digital platforms, which is a key component of our initiative. "Build Hub" seeks to create a safe and effective online platform for construction management while addressing data security and privacy issues.

3.2.6 Smart Recommendations in Real Estate Platforms

3.2.6.1 Summary of the research item

F. Thompson's study [6] is focused on the importance of intelligent suggestions in real estate platforms. It looks on the potential for personalized recommendations to enhance user experience and real estate decision-making.

3.2.6.2 Critical analysis of the research item

The advantages and disadvantages of this platform are shown in the list below:

- focuses on the importance of suggestions and personalisation in real estate platforms.
- explains the methods and algorithms used to implement intelligent suggestions.
- recognizes the potential for higher user happiness and engagement.

- Limited discussion of difficulties with adopting recommendation systems, such as worries about data privacy.
- Specifically illustrative instances of effective deployment in real estate platforms are few.

3.2.6.3 Relationship to the proposed research work

An important element of ”Build Hub” is directly influenced by Thompson’s study [6]. Our initiative intends to improve the user experience by offering tailored suggestions for building supplies and providers. ”Build Hub” addresses issues with data privacy and concentrates on usable implementation.

3.2.7 Construction Project Management Software: A Review

3.2.7.1 Summary of the research item

An overview of project management software for the construction industry is given in M. Johnson’s research article [7]. It covers numerous software options for project planning, scheduling, and resource management in the construction sector.

3.2.7.2 Critical analysis of the research item

The following list outlines the strength and weakness of this platform that we found:

- provides a thorough analysis of construction project management software.
- characteristics and capabilities of several software tools are discussed.
- demonstrates the potential for increasing project efficiency and teamwork.
- lacks a thorough examination of cutting-edge technologies like IoT and AI in building software.
- User adoption and change management issues are not adequately covered.

3.2.7.3 Relationship to the proposed research work

The landscape of construction project management software is well-explained by Johnson’s study, which is directly applicable to ”Build Hub” [7]. Our project goes beyond conventional software solutions by including cutting-edge AI-driven capabilities for project optimization and resource allocation.

3.2.8 Sustainable Construction Practices: A Review of Recent Developments

3.2.8.1 Summary of the research item

Recent advancements in environmentally friendly building techniques are reviewed in this article by A. Garcia and E. Martinez [8]. It covers techniques and innovations meant to lessen the negative effects of construction activities on the environment.

3.2.8.2 Critical analysis of the research item

The following list outlines the strength and weakness of this platform that we found:

- gives a current analysis on sustainable construction methods.
- discusses cutting-edge technology including the use of renewable energy sources and green construction materials.
- demonstrates the growing significance of sustainability in the building sector.
- There is little treatment of issues related to cost while implementing sustainable practices.
- lacks a thorough examination of the case studies highlighting effective sustainable initiatives.

3.2.8.3 Relationship to the proposed research work

The ”Build Hub” project’s overarching sustainability aims are in line with Garcia and Martinez’s study [8]. In order to motivate and enlighten people, ”Build Hub” places a strong emphasis on the usage of environmentally friendly products and procedures.

3.2.9 IoT Applications in Construction: A Comprehensive Review

3.2.9.1 Summary of the research item

J. Kim and S. Lee’s study [9] offers a thorough analysis of IoT applications in the building industry. It looks at the usage of IoT sensors and gadgets to keep an eye on and improve building operations.

3.2.9.2 Critical analysis of the research item

The following list outlines the strength and weakness of this platform that we found:

- provides a thorough overview of IoT applications for the construction industry, including a variety of use cases.
- explains the possibility for real-time observation and data-driven judgment.

- draws attention to the advantages of increased efficiency and safety.
- Limited discussion of data privacy and IoT security risks throughout building.
- lacks a thorough discussion of the difficulties involved in deploying IoT devices on building sites.

3.2.9.3 Relationship to the proposed research work

The IoT integration component of the ”Build Hub” project is directly influenced by Kim and Lee’s study [9]. The ”Build Hub” project seeks to use IoT sensors to monitor building sites in real-time, assuring efficiency and safety while resolving security and privacy issues.

3.2.10 BIM in Construction: State-of-the-Art and Future Directions

3.2.10.1 Summary of the research item

Building information modeling (BIM) is discussed in the construction industry in this study by D. Chen and Q. Wang [10]. It gives a general overview of the state of BIM technology now and some probable future developments.

3.2.10.2 Critical analysis of the research item

The following list outlines the strength and weakness of this platform that we found:

- provides information about the state of BIM technology at the moment and its uses.
- explains how BIM might improve project efficiency, collaboration, and visualization.
- highlights how BIM may help projects run more smoothly and with fewer mistakes.
- There is insufficient discussion of adoption hurdles for BIM, such as problems with training and data interoperability.
- lacks a thorough examination of BIM applications in several construction-related fields.

3.2.10.3 Relationship to the proposed research work

The BIM integration component of ”Build Hub” is directly related to Chen and Wang’s study [10]. In order to improve cooperation and visualization in building projects, our solution makes use of BIM technology. ”Build Hub” discusses BIM adoption issues and focuses on useful applications.

3.3 Literature Review Summary Table

Table 3.1: Summary of Literature Review for Development Projects

Overview of Key Findings and Insights from Literature Review on Development Projects

No.	Paper Title	Authors	Year	Summary	Relevance to Project
1	Digital Platforms in Construction Management [1]	A. Smith, B. Johnson	2021	Overview of digital platforms in construction	Foundational principles adopted
2	Construction Tech: The Digital Transformation [2]	L. Anderson	2019	Insights into tech's role in construction	Informs tech choices for "Build Hub"
3	Machine Learning in Construction Planning [3]	R. Adams, S. Patel	2022	ML's potential in improving construction efficiency	Basis for ML features in our platform
4	OCR for Document Verification in Real Estate [4]	D. Murphy	2020	Benefits of OCR in real estate document verification	Inspiration for our ID and land paper verification
5	Internet's Role in Construction Management [5]	E. White	2018	Exploration of internet-driven construction solutions	Supports our platform's online direction
6	Smart Recommendations in Real Estate Platforms[6]	F. Thompson	2021	Importance of smart recommendations in real estate	Influences our platform's recommendation system
7	User Experience in Digital Construction Platforms[7]	G. Roberts	2019	Role of UX in construction management platforms	Guides UX/UI decisions for "Build Hub"
8	Cybersecurity in Construction Management Platforms [8]	H. Lewis	2022	Emphasizing the importance of secure platforms	Informs our platform's security protocols

No.	Paper Title	Authors	Year	Summary	Relevance to Project
9	Digital Transformation in Real Estate: A Case Study [9]	I. Turner, J. Williams	2020	A deep dive into a successful digital transition	Offers potential pitfalls and success strategies
10	The Future of Construction: A Digital Forecast[10]	K. Martinez	2021	Predictions for tech's evolving role in construction	Provides a road map for future expansions of "Build Hub"

3.4 Conclusion

The review of the literature has provided insightful information about real estate research, construction management, and related fields. Every article presents important perspectives on the "Build Hub" project. We have carefully considered the benefits and drawbacks of the works in order to ensure that our project builds on their strengths while addressing their weaknesses.

Chapter 4 Software Requirement Specifications

The ”Build Hub” project’s Software Requirement Specifications (SRS) chapter explains the specific needs, functions, and design facets of the system. It makes sure that both technical and non-technical readers can understand the system’s requirements, enabling them to duplicate or improve the system based on the given information.

4.1 List of Features

List all important features of your system here.

- Safe registration and login process.
- Construction project creation, modification, deletion, and project management.
- Assign and oversee duties associated with building projects.
- Allocate resources, such as labor, equipment, and supplies, to various tasks.
- Produce thorough reports on the use of resources and project progress.
- Alert system for notifications, updates, and deadlines for tasks.
- Tools for team collaboration include chat, forums, and comment sections.
- Upload, distribute, and manage documents pertaining to construction.

4.2 Functional Requirements

The functional requirements of system are given below.

- The ability for users to securely register, log in, log out, and reset lost passwords is required. Once logged in, the system need to assign access privileges in accordance with user roles.
- New construction projects should be able to be created by authorized users, who may provide information such the project’s name, location, target completion date, and budget. Project managers and administrators.
- It is possible to set tasks for building projects, assign them to team members, and monitor their progress.
- Assign and monitor resources (people, equipment, and materials) to certain tasks or projects.
- Implement forums, chat rooms, and comment areas to allow team members to discuss specifics of

the project.

- Users should be able to safely upload, distribute, and manage contracts, drawings, and other paperwork connected to construction.
- Users ought to be informed of project updates, task due dates, and fresh communications.
- Detailed reports on project progress, resource use, and budget allocation may be generated and viewed.
- Team members may offer comments and evaluations on finished tasks and projects.
- Projects that have been completed can be archived for future use, keeping the system tidy.
- Users may browse and amend their profiles, reset their passwords, and update their contact information.
- Track expenditures and estimate expenses while managing the budget allotted to each project.
- Schedule and document quality checks, and make sure construction standards are being followed.
- Visual representation of project timelines with connected calendars for scheduling and reminders is known as timeline and calendar integration.

4.3 Quality Attributes

1. **Usability:** The system should have simple, straightforward navigation.
2. **Reliability:** The system ought to deliver reliable outcomes with less downtime.
3. **Performance:** Quick responses to queries and user activities.
4. **Maintainability:** It should be simple to upgrade and maintain the system.

4.4 Non-Functional Requirements

The non-functional criteria used to gauge the system's quality are listed in this section and include things like the system's time of availability, dependability, usability, serviceability, performance, and security.

- **Reusability:** The system's components have to be transferable to other areas of the application.
- **Performance:** The system should be able to handle 10,000 users at once.
- **Extensibility:** Additional features may be added without materially altering the current system.
- **Security:** Data encryption and prevention from unwanted access are two aspects of security.

4.5 Assumptions

The assumptions we make to develop our system are given in sub-section provided below.

- Everyone who uses a computer can utilize it.
- Daily backups of the database will be made.
- There is a reliable internet connection available to users.
- The users have basic computer literacy and familiarity with standard software interfaces.
- The hardware and software requirements for the system are within the common specifications of modern computers.
- Users have the necessary permissions and access rights to use and interact with the system as intended.

4.6 Use Cases

Table 4.1: User Login

This use case describes the process of a user logging into the system.

Name	User Login		
Actors	All registered users		
Summary	The user provides email and password on the login form, and after successful verification, redirect the user to the home page.		
Pre-Conditions	The user must be in the databases records, either added by any of the authorized users or added manually by a developer. The user must not already be logged in.		
Post-Conditions	The user's session is successfully established and shall be redirected to the home page.		
Special Requirements	None		
Basic Flow			
Actor Action		**System Response**	
1	The user opens the login page.	2	The login page is displayed asking for email and password.
2	The user enters valid email and password.	4	The system verifies the email and password, establishes a session for the user and redirects the user to the home page.
Alternative Flow			
3	The user enters invalid email or password.	4-A	The system responds with an error message: Incorrect email or password entered.

Table 4.2: User Registration

A new user registers for an account

Name	User Registration		
Actors	Guest users, Admin		
Summary	A new user registers for an account		
Pre-Conditions	The user is not already registered in the system		
Post-Conditions	A new user account is created, and the user is asked to verify their email		
Special Requirements	None		
Basic Flow			
Actor Action		**System Response**	
1	The guest user navigates to the registration page.	2	The system displays the registration form.
2	The user provides necessary details and submits.	3	The system creates a new user accounts and sends a verification email.
Alternative Flow			
2-A	The user enters an email that's already registered.	2-B	System displays an error message: "Email already registered".

Table 4.3: View Lands

Users can view lands available for purchase or rent

Name	View Lands		
Actors	Registered users		
Summary	Users can view lands available for purchase or rent		
Pre-Conditions	User is logged in		
Post-Conditions	The user views the list of available lands		
Special Requirements	None		
Basic Flow			
Actor Action		**System Response**	
1	The user selects the "View Lands" options .	2	The system displays a list of available houses.

Table 4.4: Add Land Record

Landowners can add new land records for sale or rent

Name	Add Land Record		
Actors	Landowner, Admin		
Summary	Landowners can add new land records for sale or rent		
Pre-Conditions	User is logged in and has landowner privileges		
Post-Conditions	A new land record is added to the system		
Special Requirements	None		
Basic Flow			
Actor Action		**System Response**	
1	The landowner selects the "Add Land" option.	2	The system displays an input form.
2	The landowner provides land details it	3	The system adds the new land record in the system.

Table 4.5: Edit Land Record

Landowners can edit the details of their lands

Name	Edit Land Record		
Actors	Landowner, Admin		
Summary	Landowners can edit the details of their lands		
Pre-Conditions	User is logged in and has landowner privileges		
Post-Conditions	The land record details are updated		
Special Requirements	None		
Basic Flow			
Actor Action		**System Response**	
1	The landowner selects a land records and chooses to edit.	2	The system displays the current details in editable it .
2	The landowner modifies details and submits.	3	The system updates the land record.

Table 4.6: Delete Land Record

Landowners can remove their land records from the system

Name	Delete Land Record		
Actors	Landowner, Admin		
Summary	Landowners can remove their land records from the system and perform it.		
Pre-Conditions	User is logged in and has landowner privileges		
Post-Conditions	The land record is removed from the system		
Special Requirements	None		
Basic Flow			
Actor Action		**System Response**	
1	The landowner selects a land record and chooses to delete.	2	The system confirms the deletion request.
2	The landowner confirms the deletion.	3	The system removes the land record.

Table 4.7: Sign Rental Agreement

Landowners and tenants can sign rental agreements for lands

Name	Sign Rental Agreement		
Actors	Landowner, Tenant		
Summary	Landowners and tenants can sign rental agreements for lands		
Pre-Conditions	Both users are logged in		
Post-Conditions	A rental agreement is signed and stored in the system		
Special Requirements	None		
Basic Flow			
Actor Action		**System Response**	
1	The tenant selects a land for rent and requests an agreement.	2	The system notifies the landowner.
3	The landowner agrees and provides terms.	4	The system creates a draft agreement.
5	Both parties review and sign the agreement.	6	The system finalizes and stores the agreement.

Table 4.8: Verify Land Record

Admins and verification agencies can verify the authenticity of land records

Name	Verify Land Record		
Actors	Admin, Verification Agency		
Summary	Admins and verification agencies can verify the authenticity of land record.		
Pre-Conditions	User is logged in and has verification privileges		
Post-Conditions	The land record is marked as verified in the system		
Special Requirements	None		
Basic Flow			
Actor Action		**System Response**	
1	The verifier selects a land record for verification.	2	The system displays the land details.
3	The verifier checks external sources and confirms authenticity.	4	The system marks the land record as verified.

Table 4.9: Search Lands

Users can search for lands based on various criteria

Name	Search Lands		
Actors	All users		
Summary	Users can search for lands based on various criteria and make do select this		
Pre-Conditions	None		
Post-Conditions	The user views a list of lands matching their search criteria		
Special Requirements	None		
Basic Flow			
Actor Action		**System Response**	
1	The user inputs search criteria.	2	The system displays matching land records.

Table 4.10: Leave Feedback

Users can leave feedback for lands they have interacted with

Name	Leave Feedback		
Actors	All users		
Summary	Users can leave feedback for lands they have interacted with and admin see		
Pre-Conditions	User is logged in		
Post-Conditions	Feedback is stored in the system and associated with the land record		
Special Requirements	None		
Basic Flow			
Actor Action		**System Response**	
1	The user selects a land and chooses to leave feedback.	2	The system displays a feedback form.
2	The user submits their feedback.	3	The system stores the feedback and associates it with the land record.

4.7 Hardware and Software Requirements

The following is a list of the hardware and software requirements that will be needed to build and launch the project.

4.7.1 Hardware Requirements

- **Server:** Minimum 8GB RAM, Quad-core processor, 128GB HDD.
- **Client:** Any modern computer or smartphone with a web browser.

4.7.2 Software Requirements

- **Backend:** .net.
- **Frontend:** angular.
- **Database:** SQL Server.
- **Operating System:** Linux/Windows Server.

4.8 Graphical User Interface

GUI dumps of each screen, with reference to the users and functionality are given below.

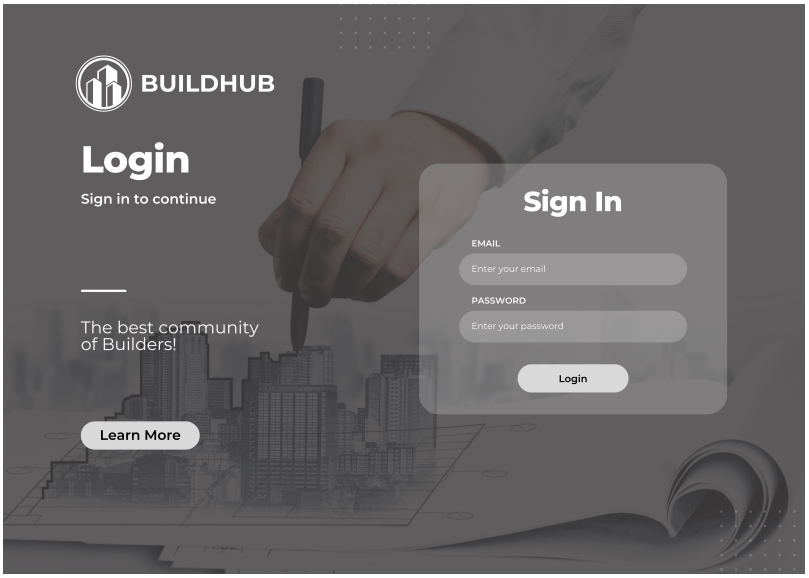


Figure 4.1: LogIn Page

Any user can login from this page

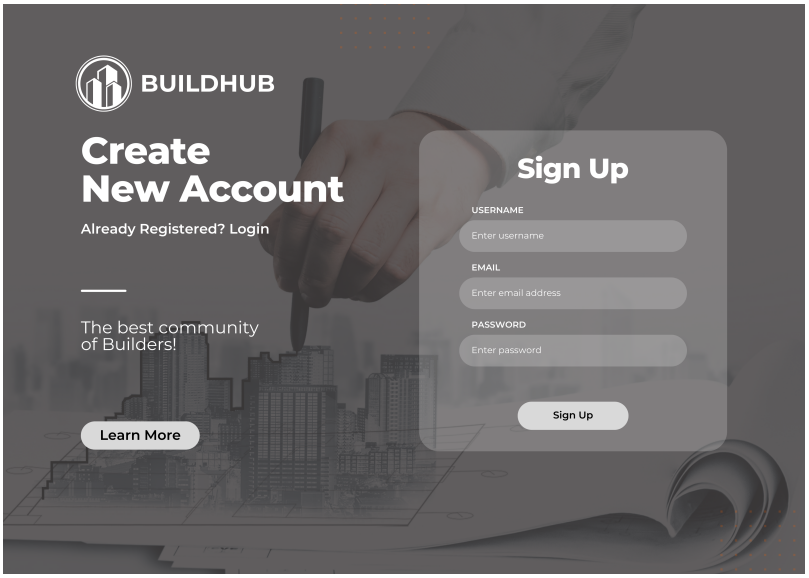


Figure 4.2: SignUp Page

Any user can SigIn and create their account

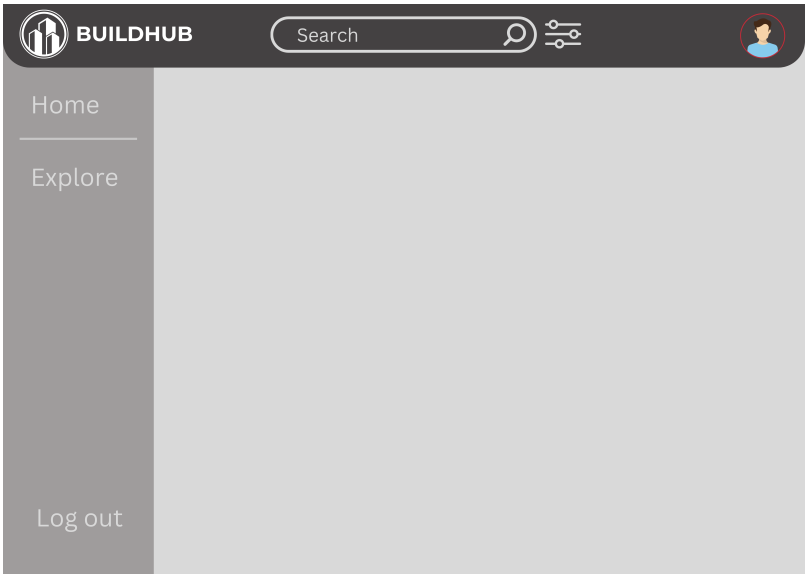


Figure 4.3: Home Page

On this page user can see his dashboard and can perform actions

4.9 Database Design

The database design of the system is provided in the sub-sections given below.

4.9.1 ER Diagram

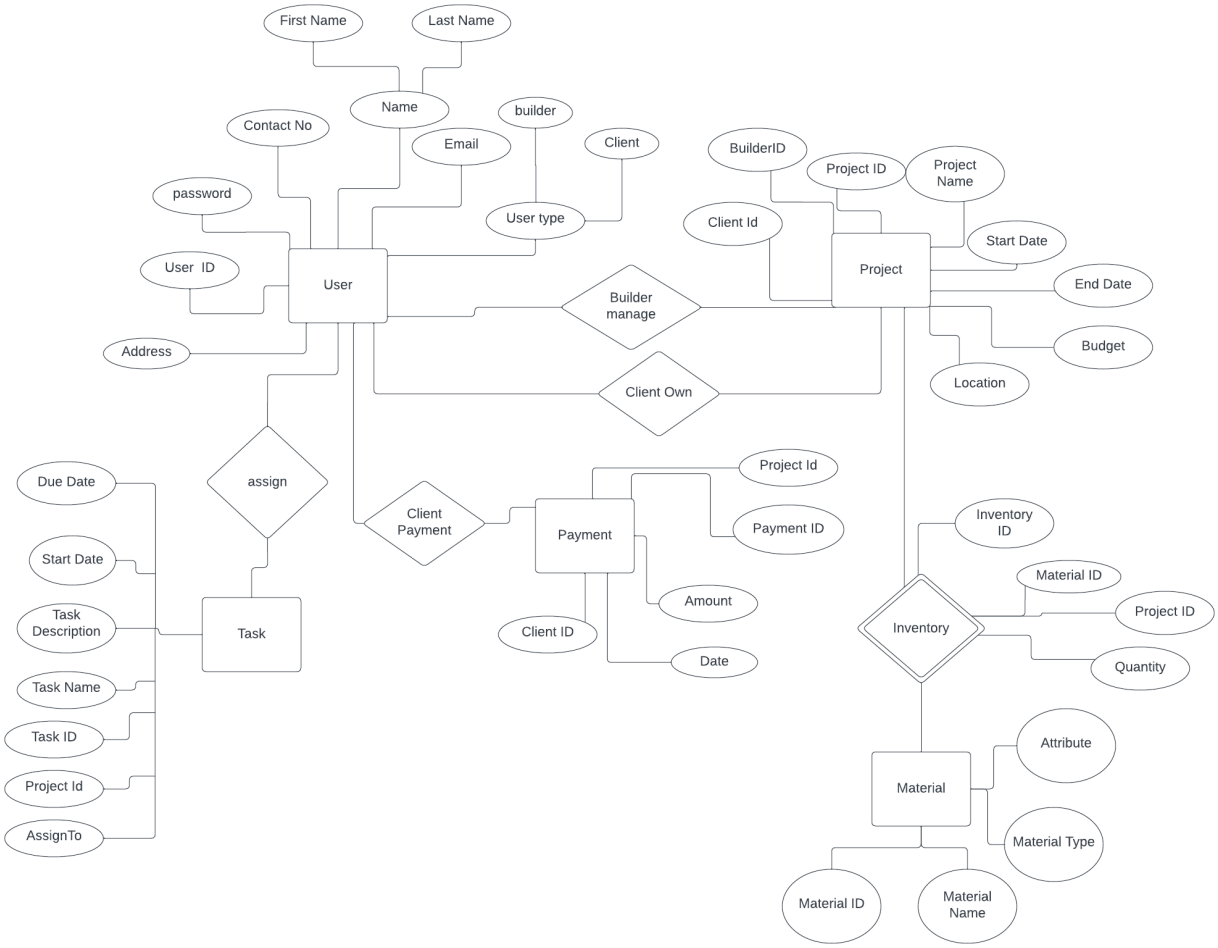


Figure 4.4: ER Diagram

This is the database design for BuildHub

4.9.2 Data Dictionary

Table 4.11: Data Dictionary for User Table

Overview of user information including ID, email, password, name, role, registration date, and last login.

Field	Type	Description
UserID	Integer	Unique identifier for each user
Email	String	Email address of the user
Password	String	Encrypted password
Name	String	Full name of the user
Role	String	Role of the user (Admin, Project Manager, Team Leader, etc.)
RegisteredDate	Date	Date when the user registered
LastLogin	DateTime	Last date and time of user login

Table 4.12: Data Dictionary for Project Table

Details of projects, such as project ID, name, start and end dates, status, and the ID of the project manager.

Field	Type	Description
ProjectID	Integer	Unique identifier for each project
Name	String	Name of the project
StartDate	Date	The start date of the project
EndDate	Date	Estimated end date of the project
Status	String	Current status (Ongoing, Completed, Paused, etc.)
ManagerID	Integer	ID of the Project Manager

Table 4.13: Data Dictionary for Agreement Table

Record of agreements specifying agreement ID, user ID, project ID, and detailed agreement information.

Field	Type	Description
AgreementID	Integer	Unique identifier for each agreement
UserID	Integer	ID of the user involved in the agreement
ProjectID	Integer	ID of the project for which the agreement is made
AgreementDetails	String	Detailed description of the agreement

Table 4.14: Data Dictionary for Land Record Table

Information on land records, including record ID, owner ID, location, area in square meters, and estimated land value.

Field	Type	Description
RecordID	Integer	Unique identifier for each land record
OwnerID	Integer	ID of the landowner
Location	String	Location of the land
Area	Float	Area of the land in square meters
Value	Float	Estimated value of the land

4.10 Risk Analysis

The risks that may be encountered during the project is listed below:

- **Technical Risks:** Integration of different parts of the angular stack, managing large datasets.
- **Business Risks:** Changing project requirements, availability of key personnel.

4.11 Conclusion

a concise explanation of expected development risks, together with the main use cases, graphical user interface design, preliminary Build Hub database design, and detailed system requirements.

Chapter 5 High-Level and Low-Level Design

The high and low level design of the Build Hub are explored in further detail in this chapter. System architecture and flow of functionalities is shown visually using diagrams. Detailed discussions on architecture, strategy, and policy are provided.

5.1 System Overview

The **Build Hub** platform aims to transform the construction industry by facilitating seamless interactions between clients and builders. Our system software comprises the following key modules:

5.1.1 Builders Profile

The system allows builders to create and manage their profiles, showcasing their qualifications, expertise, past projects, and customer reviews. Clients can explore builder profiles to make informed decisions when selecting a builder for their project.

5.1.2 Proposal Submission and Builder Verification

In this phase, builders submit their project proposals based on their expertise and availability. The system also includes builder verification to ensure their qualifications align with the project requirements.

5.1.3 Project Bidding by Builders

This module focuses on the process of project bidding by builders. It allows builders to submit their bids for projects posted by clients. Builders can specify their project execution plan, cost estimates, and timelines as part of their bids. The system facilitates bid comparison and evaluation, aiding clients in selecting the most suitable builder for their project.

5.1.4 Project Progress Tracking

This module allows clients and builders to monitor the progress of ongoing projects. The system provides tools to track milestones, make updates, and receive notifications about project milestones and deadlines.

5.1.5 Quality Assurance and Documentation

This phase involves the implementation of quality checks and documentation. It ensures that construction standards are met, and project-related documents are properly maintained and accessible to relevant

parties.

5.1.6 Cost Estimation of the Project

This subsection deals with cost estimation for construction projects. Clients and builders can use the system's tools to estimate project costs accurately. It may involve detailed cost breakdowns, material pricing, labor costs, and potential contingencies.

5.2 Design Considerations

For our application to be successful, we need to address a range of issues, assumptions, dependencies, and constraints to ensure a smooth user experience. This section outlines various considerations that need to be taken into account before devising a complete design solution.

5.2.1 Assumptions and Dependencies

The following are the assumptions and dependencies associated with our web application:

5.2.1.1 Operating System

Users must have compatible mobile devices with supported operating systems, including Android and iOS, to install and use the **Build Hub** mobile application. The mobile app will be made available for download via the Google Play Store for Android users and the Apple App Store for iOS users. The system's functionality and performance will be optimized for both Android and iOS platforms.

5.2.1.2 End-User Characteristics

- End users should have a reliable Internet connection.
- End users should possess the latest devices with up-to-date browser versions.
- End users should have basic knowledge of using the application.

5.2.2 General Constraints

Our project aims to provide an easy-to-use platform for clients and builders to interact and engage in construction projects seamlessly. However, there are certain limitations and constraints that will significantly impact the design of the system's software. These constraints include:

5.2.2.1 Hardware and Software Environment

- Internet connectivity issues.
- Overloaded website traffic, leading to potential inaccessibility for some users.
- Browser versions that are not supported.
- Security breaches by third-party software, requiring temporary shutdowns.

5.2.2.2 End-User Environment

- Users lacking basic internet knowledge.
- Users lacking understanding of how to use web applications.

5.2.2.3 Availability or Volatility of Resources

- Poor Internet connections.
- Outdated or underpowered devices.
- Slow loading times due to insufficient speed.

5.2.2.4 Interoperability Requirements

- Ensuring smooth connections between users and the database.

5.2.2.5 Interface/Protocol Environments

- User-friendly interfaces for easy navigation within the system.

5.2.2.6 Data Repository and Distribution Requirements

- Securely storing user information to prevent data breaches.
- Ensuring the database's security to avoid data leaks.

5.2.2.7 Performance Requirements

- The system must efficiently handle a large number of users concurrently.
- The system should accurately send notifications and alerts to users about project updates.
- The system's database must remain operational without causing issues.

5.2.3 Goals and Guidelines

The design of the **Build Hub** platform should align with specific goals and guidelines to achieve its intended purpose:

5.2.3.1 Efficiency

- The system should streamline interactions between clients and builders, ensuring efficient project management and communication.

5.2.3.2 Transparency

- Transparency in the builder-client interaction is essential, making it easier for clients to make informed decisions.

5.2.3.3 Security

- Implement advanced technology, like machine learning and OCR verification, to ensure secure and trustworthy processes.

5.2.3.4 User Experience

- Prioritize an intuitive user interface and user experience to enhance user satisfaction.

5.2.4 Development Methods

In the development of the **Build Hub** platform, we will employ the Scrum methodology, an agile development approach known for its iterative and repeatable cycles. Work will be divided into several sprints, each with set deadlines not exceeding one week. After each sprint, a 15-to-20-minute meeting will be held to review progress and make necessary schedule adjustments. Scrum's adaptability allows for changes in projects or the addition of new functionalities, making it a suitable choice for our development methodology.

5.3 System Architecture

The project bidding mobile application offers a robust architecture designed to support seamless interactions between clients and builders. This architecture leverages Dart for both frontend and backend development, ensuring a responsive and efficient mobile application. SQL Server serves as the database, offering real-time data storage and synchronization for enhanced user experiences.

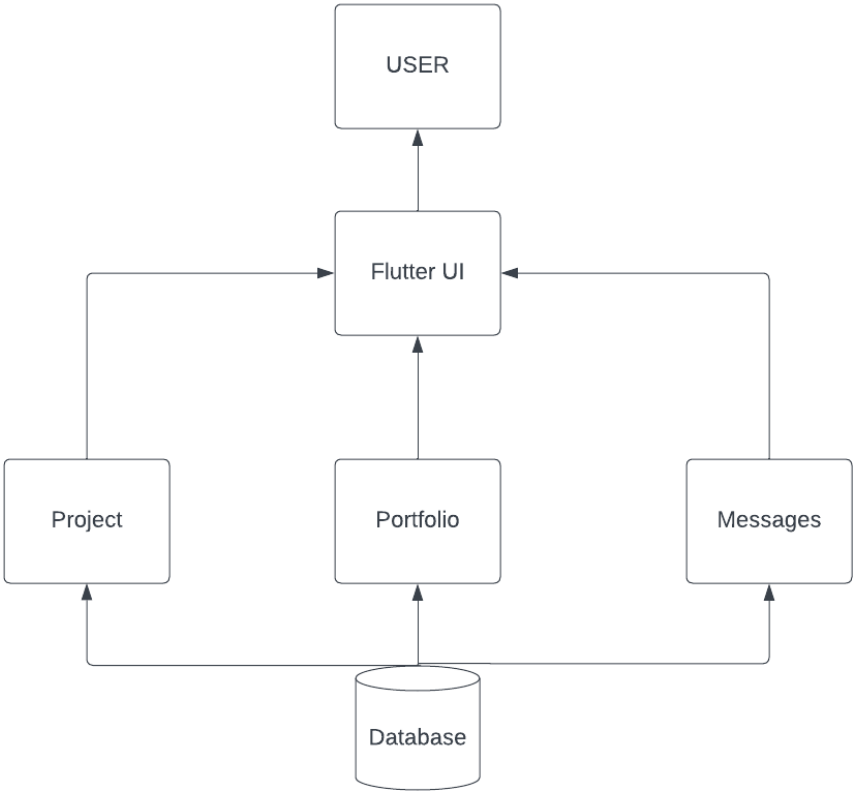


Figure 5.1: System Architecture diagram

This is the high level system Architecture Diagram

5.3.1 Project Matching and Posting Module

At the core of the system architecture is the Project Matching and Posting Module. This component allows clients to post construction projects and builders to express their interest. Utilizing Dart, the frontend interface provides a user-friendly environment for posting projects and viewing builder proposals. The backend processes and manages these project details, creating a seamless connection between clients and builders.

5.3.2 Proposal Submission and Builder Verification Module

Built with Dart, the Proposal Submission and Builder Verification Module is dedicated to proposal submission by builders. Builders can submit proposals based on their qualifications and availability. The backend, utilizing Dart as well, handles builder verification, enhancing trust and transparency by ensuring that builders' qualifications align with

5.3.3 Project Bidding by Builders Module

Efficient project bidding is vital to the system's architecture. This module, designed in Dart, allows builders to submit bids for available projects, specifying their project plans, cost estimates, and timelines. The backend, also implemented in Dart, facilitates bid comparison and evaluation, providing clients with the data needed to make informed decisions.

5.3.4 Project Progress Tracking Module

Real-time monitoring of project progress is a key feature of the system architecture. Utilizing Dart, the frontend interface empowers clients and builders with tools to track project milestones, make updates, and receive notifications about project progress. The backend, also developed in Dart, handles the synchronization and storage of project data in SQL Server.

5.3.5 Quality Assurance and Documentation Module

Quality assurance and documentation are integral to the construction process. This module ensures that construction standards are met, and all project-related documents are accurately maintained. The Dart-based frontend provides a user-friendly interface, while SQL Server's real-time data storage ensures data accuracy and accessibility.

5.3.6 Cost Estimation of the Project Module

Cost estimation is a critical component of construction projects. The frontend and backend, both powered by Dart, include a module that allows clients and builders to estimate project costs accurately, with detailed breakdowns, material pricing, labor costs, and contingencies. SQL Server stores and syncs this data in real-time, providing accurate cost estimates for all stakeholders.

5.3.7 Server

The server component, powered by Dart, serves as the backbone of the architecture. It orchestrates all system operations, handles user inputs, communicates with SQL Server, and controls data flow within the application. Dart's versatility makes it an ideal choice for the server, ensuring seamless coordination between the frontend, backend, and database components.

5.4 Architectural Strategies

The **Build Hub** project adopts the following architectural strategies to ensure the success of the platform:

5.4.1 Programming Language

The primary programming language chosen for this project is Dart. Dart is a versatile language that allows us to develop both the frontend and backend components, ensuring a consistent codebase and ease of maintenance. Dart's compatibility with the angular framework, a leading choice for mobile app development, makes it an ideal language for this project.

5.4.2 Product Enhancement and Extensibility

The system's architecture is designed with extensibility in mind. It can be easily extended to add more functionalities for user interaction. Potential enhancements may include features like job rating, improved project search, and expanded user profiles. These extensions can be implemented with ease, thanks to the flexibility of Dart, allowing the platform to evolve to meet future requirements.

5.4.3 Concurrency and Synchronization

This platform utilizes several strategies to ensure smooth concurrency and synchronization of operations:

- Async operations will be used on the frontend, so the user is able to interact with the mobile application while the system is working on any time-consuming operations like web API requests.
- Parallel Processing on Backend: The backend application incorporates parallel processing to adeptly manage multiple users simultaneously. This ensures efficient handling of concurrent requests, preventing bottlenecks and optimizing overall system performance during peak usage times.
- Caching Mechanism: Implementing a caching mechanism helps enhance system performance by storing frequently accessed data temporarily. This reduces the need for redundant computations and database queries, resulting in faster response times for users and reduced server load.
- Load Balancing: To distribute incoming network traffic evenly across multiple servers, load balancing is employed. This not only prevents any single server from becoming a point of failure but also ensures that resources are utilized efficiently, contributing to a more reliable and scalable system architecture.

5.5 Domain Model/Class Diagram

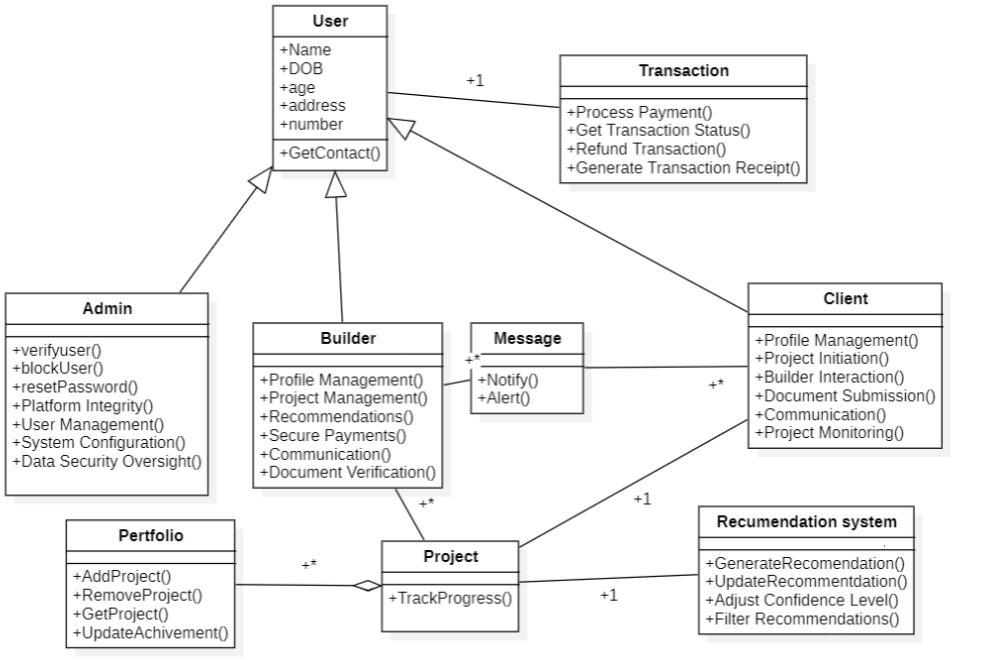


Figure 5.2: Class Diagram

This is the Class Diagram for BuildHub

5.6 Sequence Diagrams

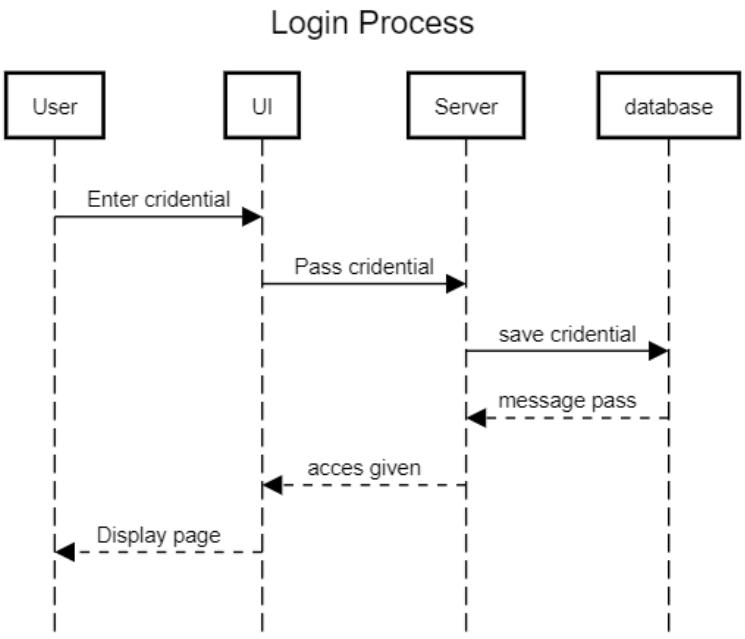


Figure 5.3: Login Process

Illustrates the sequence of steps involved in user authentication and login

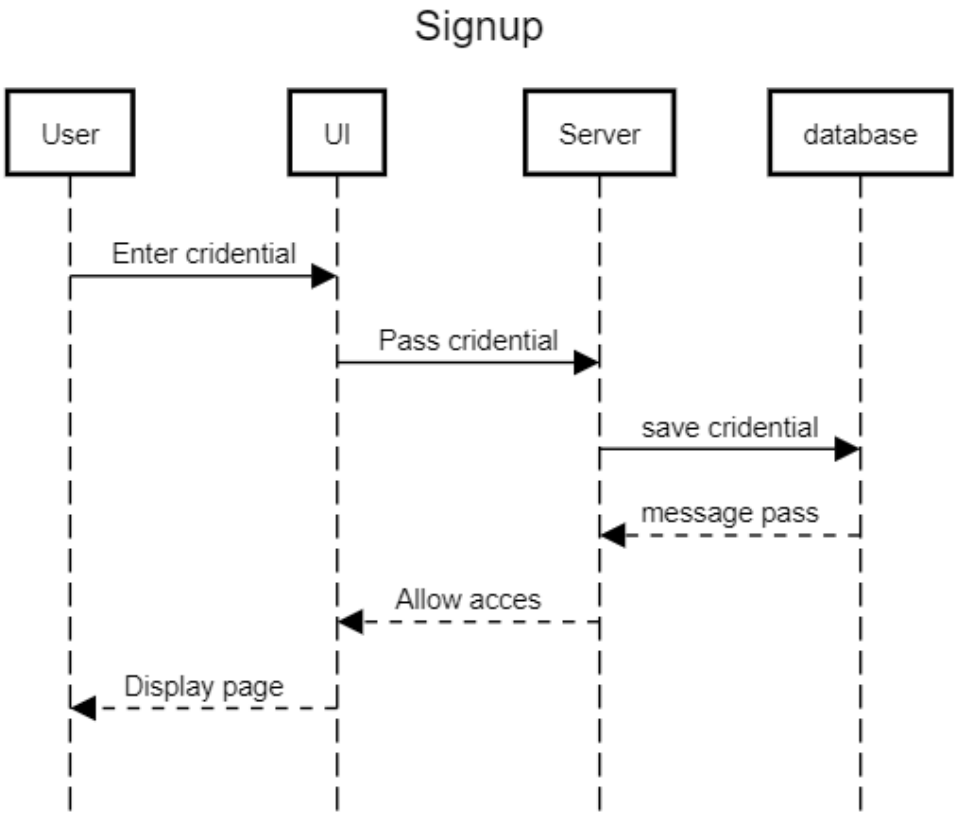


Figure 5.4: Sign-Up Procedure

Demonstrates the flow of actions for user registration and account creation

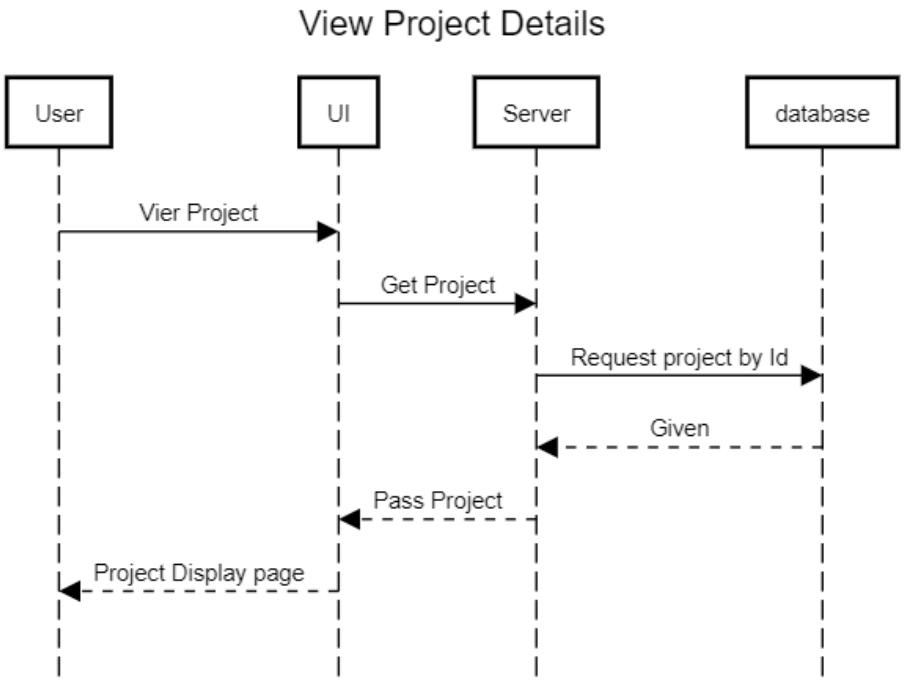


Figure 5.5: View Project Details

Presents the sequence of events when a user accesses and views project information

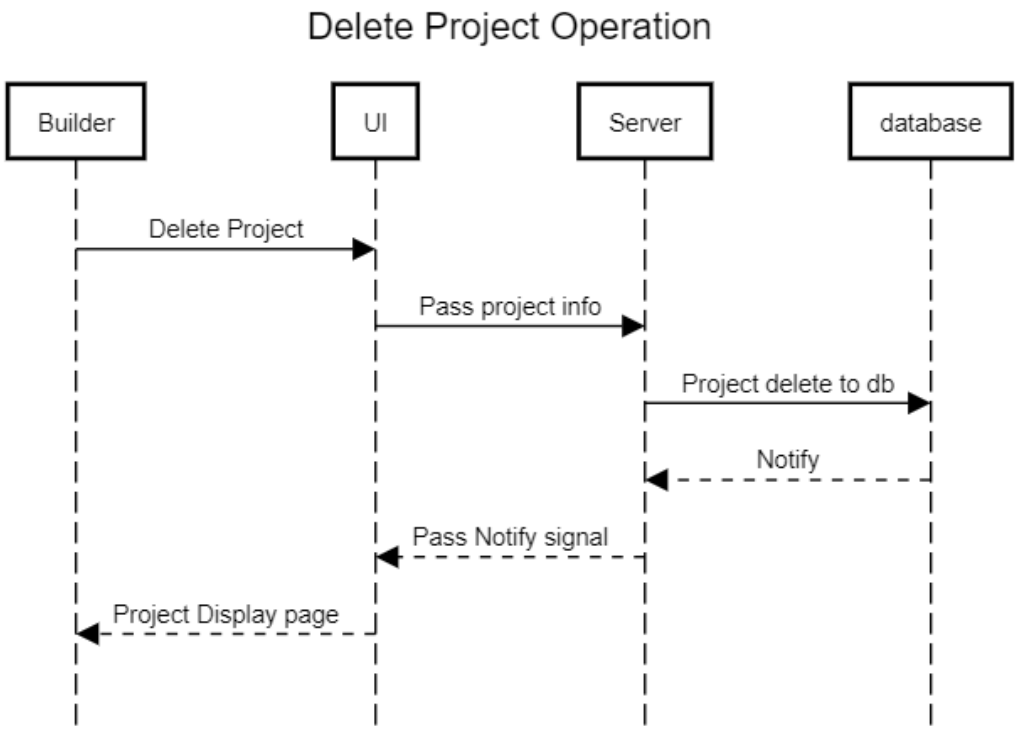


Figure 5.6: Delete Project Operation

Depicts the process of deleting a project from the system, including necessary confirmations

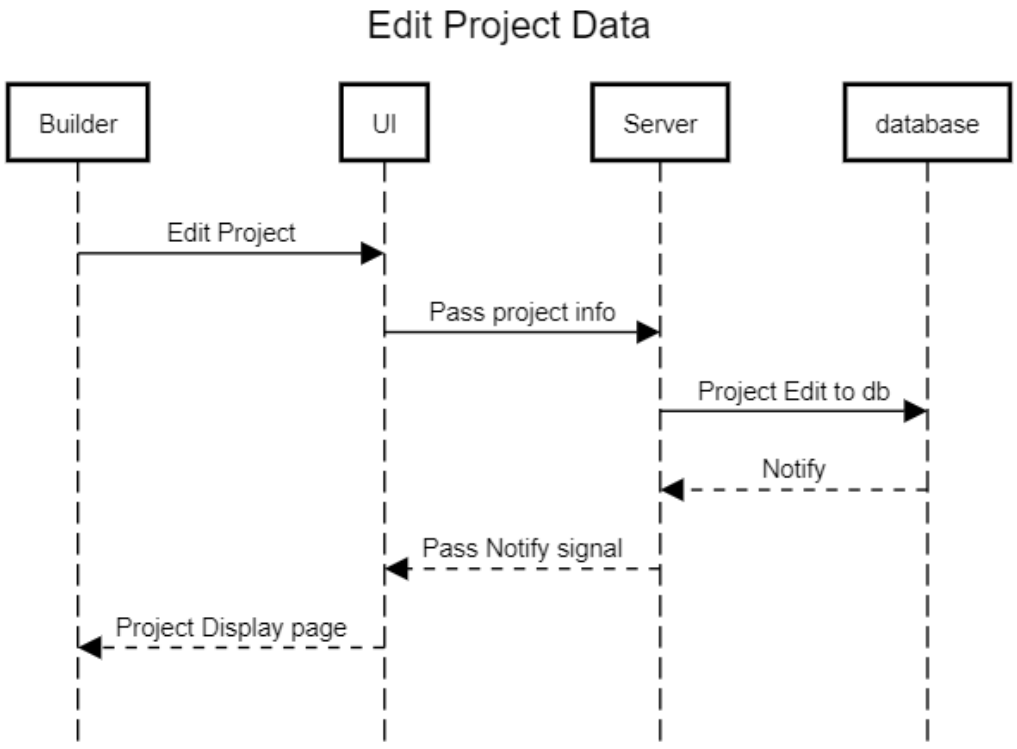


Figure 5.7: Edit Project Data

Illustrates the steps involved in modifying project details, showcasing the update workflow

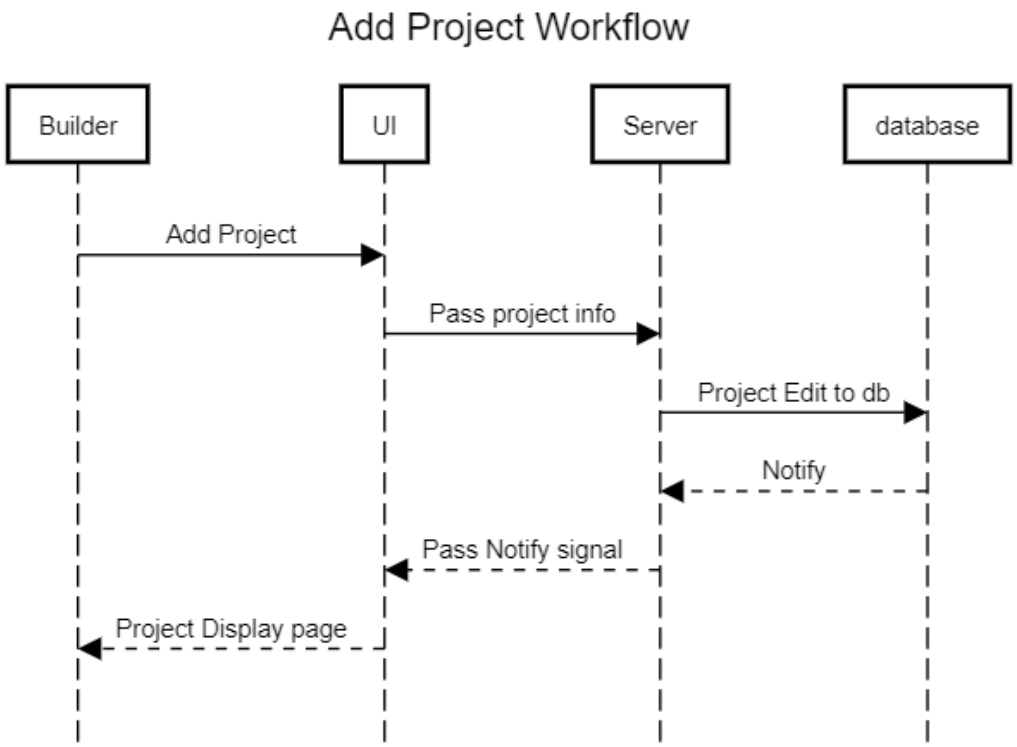


Figure 5.8: Add Project Workflow

Shows the sequence of actions for adding a new project to the system, including data input and validation

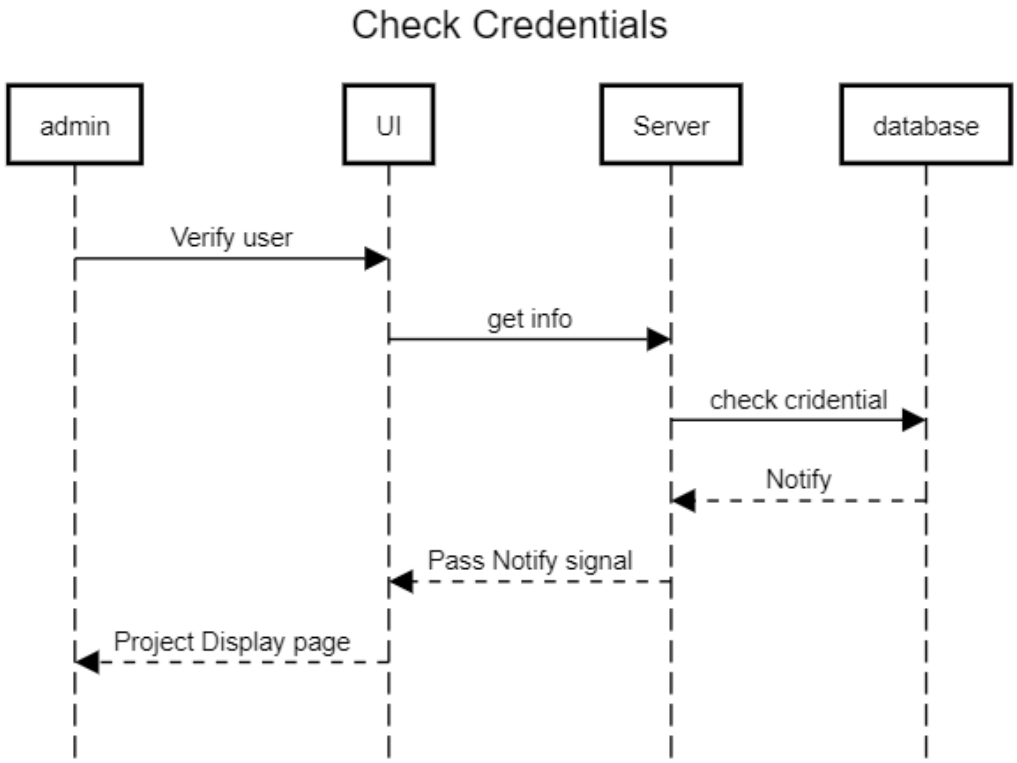


Figure 5.9: Check Credentials

Details the sequence of events involved in verifying user credentials during the authentication process

5.7 Policies and Tactics

The following considerations are essential for the project's success:

5.7.1 Tools to be Used

Several APIs, tools, and external libraries are integral to the **Build Hub** project:

- Dart (Programming language for mobile application)
- angular (Open-source UI software development toolkit)
- SQL Server (Remote database server)

5.7.2 Coding Guidelines and Conventions

Dart is a selected programming language for the project. The project adheres to stringent coding guidelines and conventions, ensuring code consistency and maintainability.

5.7.3 User Interface

This application features a modern and straightforward user interface (UI) designed to provide an appealing and user-friendly experience. The UI is characterized by simplicity, relevant color schemes, and an intuitive design, ensuring that users can easily navigate and interact with the platform.

5.7.4 Extensibility

The project considers extensibility as a crucial aspect of its design. To accommodate future feature additions and the scaling up of the user base, a modular approach is employed. The development includes the creation of microservices for various components of the project, enhancing integration capabilities and allowing for seamless expansion in the future.

5.7.5 Policies for System Testing

The **Build Hub** project employs a comprehensive testing approach, which includes both white box and black box testing. Before deploying the application, every aspect is rigorously tested, ensuring that the platform functions smoothly and without errors.

5.7.6 Software Maintenance

The commitment to the **Build Hub** project extends beyond its initial development. Future feature expansions and system maintenance are dependent on the availability of the project's advisor and allocated

time resources. Upgraded features will be implemented as the project progresses.

5.7.7 Algorithms

Machine learning algorithms are integral to the project's success. Our platform relies on data analysis and processing from builders' data to facilitate efficient interactions between clients and builders. These algorithms are essential for various tasks within our system, such as shortlisting builders based on their proposals, analyzing project details, or detecting potential issues during the construction process. By applying machine learning techniques, we can streamline project matching, proposal evaluation, and project progress monitoring, ensuring a seamless experience for our users.

5.8 Conclusion

With the help of an intuitive mobile app driven by SQL Server and Dart, the "Build Hub" platform revolutionizes construction collaboration. Important aspects including builder profiles, project proposals, bidding, progress tracking, quality assurance, and cost estimation are covered by its modular design. With agility development and openness as top priorities, the platform uses Scrum to provide flexibility. "Build Hub" strives to simplify construction project management with a modern UI and machine learning integration, emphasizing user pleasure and ongoing progress.

Chapter 6 Implementation and Test Cases

Technical specifics detail are included in the overall high-level design are covered in the following part, along with discussions about the project's implementation thus far. Since the project is still in its early stages, the content might change.

6.1 Implementation

The next section provides the specific information on the implementation specifics of the components and algorithms of the designed prototype.

6.1.1 Login/SignUp

As the first point of contact for users, the "Build Hub" platform's Login/SignUp module allows for safe authentication and user account creation. To offer a smooth and simple user experience, the solution makes advantage of the angular framework. Important elements and features comprise:

- The use of angular in the authentication process ensures safe and effective login functionality. The user's credentials are safely sent to the backend for verification.
- Through the Sign-Up procedure, new users can register for accounts on the website. When creating an account, angular records user input and sends it safely to the backend. To improve security, passwords are hashed.
- The.NET-implemented backend securely stores user data, including login credentials. This guarantees data integrity and permits scalability in the future.

6.1.2 Prototype description

The angular frontend framework is used in the development of the "Build Hub" prototype, which offers a cross-platform user experience. The prototype's design prioritizes providing a good User Experience (UX) while being aesthetically pleasing and easy to use. The prototype has the following features:

- Using angular, developers can create a single codebase that functions flawlessly on both iOS and Android smartphones, giving consumers a consistent experience.
- The User Interface (UI) is designed with visually attractive design components and simple navigation to improve usability.
- The prototype is made to adjust to multiple screen sizes so that users on different devices may have a consistent and interesting experience.

6.1.3 Backend

The ”Build Hub” platform's backend is built on the.NET framework, which offers a stable and expandable basis for business logic and data storage. Important facets of the backend implementation consist of:

- The backend database safely stores user data, including login credentials. Retrieving and managing data effectively is made possible by the.NET framework.
- To facilitate smooth connection between the frontend and backend components, the backend offers APIs (Application Programming Interfaces). This guarantees updates and data interchange in real time.
- To protect user data and guarantee the integrity of the platform, security measures are put in place. Examples of these include encryption for data transfer and secure storage of sensitive information.

6.2 Test Case Design and Description

6.2.1 Client Side Test Cases

6.2.1.1 Login

Table 6.1: Login Test Case

Login			
Test Case ID:	*1*	**QA Test Engineer:**	*Ali Hamza*
Test case Version:	*1*	**Reviewed By:**	*Hassan Khalid*
Test Date:	*25/4/2024*	**Use Case Reference(s):**	*Login use case*
Revision History:	*None*		
Objective:	*The user shall be able to login to the application*		
Product/Ver/ Module:	*User Controller.*		
Environment:	*.Net 6.0 with up-to-date OS.*		
Assumptions:	*App has a stable internet connection*		
Pre-Requisite:	*Sign-up*		
Step No.	**Execution description**	**Procedure result**	
	The user clicks on the login button	*Login page displays*	
	User enters credentials and press login	*The user gets logged into the app.*	
Comments: The test case is passed. Our system is working according to our document requirement.			
Passed			

6.2.1.2 SignUp

Table 6.2: Sign Up Test Case

Sign Up			
Test Case ID:	*2*	**QA Test Engineer:**	*Ali Hamza*
Test case Version:	*1*	**Reviewed By:**	*Hassan Khalid*
Test Date:	*25/4/2024*	**Use Case Reference(s):**	*Sign-up use case*
Revision History:	*None*		
Objective:	*The user shall be able to signup to the application*		
Product/Ver/ Module:	*User Controller.*		
Environment:	*.Net 6.0 with up-to-date OS.*		
Assumptions:	*Web has a stable internet connection*		
Pre-Requisite:	*None.*		
Step No.	**Execution description**	**Procedure result**	
	The user clicks on the signup button	*SignUp page displays*	
	The user fills details and clicks signup	*The user gets signed up into the Web.*	
Comments: The test case is passed. Our system is working according to our document requirement.			
Passed			

6.2.1.3 Update Profile

Table 6.3: Update Profile Test Case

Update Profile			
Test Case ID:	*3*	**QA Test Engineer:**	*Ali Hamza*
Test case Version:	*1*	**Reviewed By:**	*Hassan Khalid*
Test Date:	*25/4/2024*	**Use Case Reference(s):**	*Profile use case*
Revision History:	*None*		
Objective:	*The user shall be able to Update Profile*		
Product/Ver/ Module:	*User Controller.*		
Environment:	*.Net 6.0 with up-to-date OS.*		
Assumptions:	*Web has a stable internet connection*		
Pre-Requisite:	*None.*		
Step No.	**Execution description**	**Procedure result**	
	The user clicks on the Update Profile	*Edit Profile page displays*	
	The user update details	*The user gets Success message*	
Comments: The test case is passed. Our system is working according to our document requirement.			
Passed			

6.2.1.4 View Profile

Table 6.4: View Profile Test Case

View Profile			
Test Case ID:	*4*	**QA Test Engineer:**	*Ali Hamza*
Test case Version:	*1*	**Reviewed By:**	*Hassan Khalid*
Test Date:	*25/4/2024*	**Use Case Reference(s):**	*View Profile use case*
Revision History:	*None*		
Objective:	*The user shall be able to View Profile*		
Product/Ver/ Module:	*User Controller.*		
Environment:	*.Net 6.0 with up-to-date OS.*		
Assumptions:	*Web has a stable internet connection*		
Pre-Requisite:	*None.*		
Step No.	**Execution description**	**Procedure result**	
	The user clicks on the View Profile	*View Profile page displays*	
Comments: The test case is passed. Our system is working according to our document requirement.			
Passed			

6.2.1.5 Forget Password

Table 6.5: Forget Password Test Case

Forget Password			
Test Case ID:	*5*	**QA Test Engineer:**	*Ali Hamza*
Test case Version:	*1*	**Reviewed By:**	*Hassan Khalid*
Test Date:	*25/4/2024*	**Use Case Reference(s):**	*Password use case*
Revision History:	*None*		
Objective:	*The user shall be able to Forget Password*		
Product/Ver/ Module:	*User Controller.*		
Environment:	*.Net 6.0 with up-to-date OS.*		
Assumptions:	*Web has a stable internet connection*		
Pre-Requisite:	*None.*		
Step No.	**Execution description**	**Procedure result**	
	The user clicks on the Forget Password	*Forget Password page displays*	
	The user enter mail	*The user gets Email with code*	
Comments: The test case is passed. Our system is working according to our document requirement.			
Passed			

6.2.1.6 Search Builder

Table 6.6: Search Builder Test Case

Search Builder			
Test Case ID:	*6*	**QA Test Engineer:**	*Ali Hamza*
Test case Version:	*1*	**Reviewed By:**	*Hassan Khalid*
Test Date:	*25/4/2024*	**Use Case Reference(s):**	*Builder use case*
Revision History:	*None*		
Objective:	*Verify that users can effectively use the Search Builder feature.*		
Product/Ver/ Module:	*Search Module*		
Environment:	*Web-based application with stable internet connection.*		
Assumptions:	*Users have necessary permissions to access Search Builder.*		
Pre-Requisite:	*Logged into the application.*		
Step No.	**Execution description**	**Procedure result**	
	Navigate to the Search Builder feature.	*Search Builder interface loads.*	
	Add search criteria and apply filters.	*Relevant search results are displayed.*	
Comments: The test case is passed. Our system is working according to our document requirement.			
Passed			

6.2.1.7 View Builder

Table 6.7: View Builder Test Case

View Builder			
Test Case ID:	*7*	**QA Test Engineer:**	*Ali Hamza*
Test case Version:	*1*	**Reviewed By:**	*Hassan Khalid*
Test Date:	*25/4/2024*	**Use Case Reference(s):**	*View Builder use case*
Revision History:	*None*		
Objective:	*Ensure that users can effectively create views using the View Builder feature.*		
Product/Ver/ Module:	*View Module*		
Environment:	*Web-based application with stable internet connection.*		
Assumptions:	*Users have necessary permissions to access View Builder.*		
Pre-Requisite:	*Logged into the application.*		
Step No.	**Execution description**	**Procedure result**	
	Access the View Builder feature.	*View Builder interface loads successfully.*	
	Create new view with components	*Can be accessed for data visualization.*	
Comments: The test case is passed. Our system is working according to our document requirement.			
Passed			

6.2.1.8 Add Project

Table 6.8: Add Project Test Case

Add Project			
Test Case ID:	*8*	**QA Test Engineer:**	*Ali Hamza*
Test case Version:	*1*	**Reviewed By:**	*Hassan Khalid*
Test Date:	*25/4/2024*	**Use Case Reference(s):**	*Project use case*
Revision History:	*None*		
Objective:	*Verify that users can successfully add a new project.*		
Product/Ver/ Module:	*Project Management Module*		
Environment:	*Web-based application with stable internet connection.*		
Assumptions:	*User has appropriate permissions to add projects.*		
Pre-Requisite:	*Logged into the application.*		
Step No.	**Execution description**	**Procedure result**	
	Navigate to the 'Add Project' section.	*'Add Project' page is accessible.*	
	Fill in project details.	*Project details are entered.*	
	Click on the 'Save' button.	*Project is saved in the system.*	
	Verify the project appears in the project list.	*Project is listed among projects.*	
Comments: The test case is passed. Our system is working according to our document requirement.			
Passed			

6.2.1.9 Get Projects

Table 6.9: Get Projects Test Case

Get Projects			
Test Case ID:	*9*	**QA Test Engineer:**	*Ali Hamza*
Test case Version:	*1*	**Reviewed By:**	*Hassan Khalid*
Test Date:	*25/4/2024*	**Use Case Reference(s):**	*Get Projects use case*
Revision History:	*None*		
Objective:	*Ensure that users can retrieve the list of projects from the system.*		
Product/Ver/ Module:	*Project Management Module*		
Environment:	*Web-based application with stable internet connection.*		
Assumptions:	*User has appropriate permissions to access projects.*		
Pre-Requisite:	*Logged into the application.*		
Step No.	**Execution description**	**Procedure result**	
	Navigate to the 'Projects' section.	*'Projects' page is accessible.*	
	Retrieve the list of projects.	*List of projects is successfully fetched.*	
	Verify the presence of expected projects.	*Expected projects are listed*	
Comments: The test case is passed. Our system is working according to our document requirement.			
Passed			

6.2.1.10 View Project

Table 6.10: View Project Test Case

View Project			
Test Case ID:	*10*	**QA Test Engineer:**	*Ali Hamza*
Test case Version:	*1*	**Reviewed By:**	*Hassan Khalid*
Test Date:	*25/4/2024*	**Use Case Reference(s):**	*View Project use case*
Revision History:	*None*		
Objective:	*Verify that users can view project details.*		
Product/Ver/ Module:	*Project Management Module*		
Environment:	*Web-based application with stable internet connection.*		
Assumptions:	*User has appropriate permissions to view project details.*		
Pre-Requisite:	*Logged into the application and project exists.*		
Step No.	**Execution description**	**Procedure result**	
	Navigate to the 'Projects' section.	*'Projects' page is accessible.*	
	Select the project to view.	*Selected project details are displayed.*	
Comments: The test case is passed. Our system is working according to our document requirement.			
Passed			

6.2.1.11 Get Messages

Table 6.11: Get Messages Test Case

Get Messages			
Test Case ID:	*11*	**QA Test Engineer:**	*Ali Hamza*
Test case Version:	*1*	**Reviewed By:**	*Hassan Khalid*
Test Date:	*25/4/2024*	**Use Case Reference(s):**	*Messages use case*
Revision History:	*None*		
Objective:	*Verify that users can retrieve messages from the system.*		
Product/Ver/ Module:	*Messaging Module*		
Environment:	*Web-based application with stable internet connection.*		
Assumptions:	*User has appropriate permissions to access messages.*		
Pre-Requisite:	*Logged into the application.*		
Step No.	**Execution description**	**Procedure result**	
	Navigate to the 'Messages' section.	*'Messages' page is accessible.*	
	Retrieve the list of messages.	*List of messages is successfully fetched.*	
	Verify the presence of expected messages	*Expected messages are listed.*	
Comments: The test case is passed. Our system is working according to our document requirement.			
Passed			

6.2.1.12 View Message

Table 6.12: View Message Test Case

View Message			
Test Case ID:	*12*	**QA Test Engineer:**	*Ali Hamza*
Test case Version:	*1*	**Reviewed By:**	*Hassan Khalid*
Test Date:	*25/4/2024*	**Use Case Reference(s):**	*View Message use case*
Revision History:	*None*		
Objective:	*Verify that users can view a specific message.*		
Product/Ver/ Module:	*Messaging Module*		
Environment:	*Web-based application with stable internet connection.*		
Assumptions:	*User has appropriate permissions to view messages.*		
Pre-Requisite:	*Logged into the application and message exists.*		
Step No.	**Execution description**	**Procedure result**	
	Navigate to the 'Messages' section.	*'Messages' page is accessible.*	
	Select the specific message to view.	*Selected message details are displayed.*	
Comments: The test case is passed. Our system is working according to our document requirement.			
Passed			

6.2.1.13 Add Bidding

Table 6.13: Add Bidding Test Case

Add Bidding			
Test Case ID:	*13*	**QA Test Engineer:**	*Ali Hamza*
Test case Version:	*1*	**Reviewed By:**	*Hassan Khalid*
Test Date:	*25/4/2024*	**Use Case Reference(s):**	*Add Bidding use case*
Revision History:	*None*		
Objective:	*Verify that users can add a new bidding entry.*		
Product/Ver/ Module:	*Bidding Module*		
Environment:	*Web-based application with stable internet connection.*		
Assumptions:	*User has appropriate permissions to add bidding entries.*		
Pre-Requisite:	*Logged into the application and project exists.*		
Step No.	**Execution description**	**Procedure result**	
	Navigate to the 'Bidding' section.	*'Bidding' section is accessible.*	
	Click on the 'Add Bidding' button.	*Bidding form is displayed.*	
	Fill in the necessary details.	*Bidding details are entered successfully.*	
	Click on the 'Submit' button.	*New bidding entry is added to the system.*	
Comments: The test case is passed. Our system is working according to our document requirement.			
Passed			

6.2.2 Builder Side Test Cases

6.2.2.1 Get Clients

Table 6.14: Get Clients Test Case

Get Clients			
Test Case ID:	*1*	**QA Test Engineer:**	*Ali Hamza*
Test case Version:	*1*	**Reviewed By:**	*Hassan Khalid*
Test Date:	*25/4/2024*	**Use Case Reference(s):**	*Get Clients use case*
Revision History:	*None*		
Objective:	*Verify that builders can retrieve clients' information.*		
Product/Ver/ Module:	*Client Management Module*		
Environment:	*Web-based application with stable internet connection.*		
Assumptions:	*Builder has appropriate permissions to access clients' information.*		
Pre-Requisite:	*Logged into the application.*		
Step No.	**Execution description**	**Procedure result**	
	Navigate to the 'Clients' section.	*'Clients' page is accessible.*	
	Retrieve the list of clients.	*List of clients is successfully fetched.*	
	Verify the presence of expected clients.	*Expected clients are listed.*	
Comments: The test case is passed. Our system is working according to our document requirement.			
Passed			

6.2.2.2 View Client

Table 6.15: View Client Test Case

View Client			
Test Case ID:	*2*	**QA Test Engineer:**	*Ali Hamza*
Test case Version:	*1*	**Reviewed By:**	*Hassan Khalid*
Test Date:	*25/4/2024*	**Use Case Reference(s):**	*View Client use case*
Revision History:	*None*		
Objective:	*Verify that builders can view detailed information about a client.*		
Product/Ver/ Module:	*Client Management Module*		
Environment:	*Web-based application with stable internet connection.*		
Assumptions:	*Builder has appropriate permissions to view client details.*		
Pre-Requisite:	*Logged into the application and client exists.*		
Step No.	**Execution description**	**Procedure result**	
	Navigate to the 'Clients' section.	*'Clients' page is accessible.*	
	Select the specific client to view details.	*Client details page is displayed.*	
Comments: The test case is passed. Our system is working according to our document requirement.			
Passed			

6.2.2.3 Get Notifications

Table 6.16: Get Notifications Test Case

Get Notifications			
Test Case ID:	*3*	**QA Test Engineer:**	*Ali Hamza*
Test case Version:	*1*	**Reviewed By:**	*Hassan Khalid*
Test Date:	*25/4/2024*	**Use Case Reference(s):**	*Notifications use case*
Revision History:	*None*		
Objective:	*Verify that users can retrieve notifications.*		
Product/Ver/ Module:	*Notification Management Module*		
Environment:	*Web-based application with stable internet connection.*		
Assumptions:	*User has appropriate permissions to access notifications.*		
Pre-Requisite:	*Logged into the application.*		
Step No.	**Execution description**	**Procedure result**	
	Navigate to the 'Notifications' section.	*'Notifications' page is accessible.*	
	Retrieve the list of notifications.	*List of notifications is successfully fetched.*	
Comments: The test case is passed. Our system is working according to our document requirement.			
Passed			

6.2.2.4 View Notification

Table 6.17: View Notification Test Case

View Notification			
Test Case ID:	*4*	**QA Test Engineer:**	*Ali Hamza*
Test case Version:	*1*	**Reviewed By:**	*Hassan Khalid*
Test Date:	*25/4/2024*	**Use Case Reference(s):**	*Notify use case*
Revision History:	*None*		
Objective:	*Verify that users can view detailed information about a notification.*		
Product/Ver/ Module:	*Notification Management Module*		
Environment:	*Web-based application with stable internet connection.*		
Assumptions:	*User has appropriate permissions to view notification details.*		
Pre-Requisite:	*Logged into the application and notification exists.*		
Step No.	**Execution description**	**Procedure result**	
	Navigate to the 'Notifications' section.	*'Notifications' page is accessible.*	
	Select the specific notification to view details.	*Notification details page is displayed.*	
Comments: The test case is passed. Our system is working according to our document requirement.			
Passed			

6.2.2.5 Check Project Progress

Table 6.18: Check Project Progress Test Case

Check Project Progress			
Test Case ID:	*5*	**QA Test Engineer:**	*Ali Hamza*
Test case Version:	*1*	**Reviewed By:**	*Hassan Khalid*
Test Date:	*25/4/2024*	**Use Case Reference(s):**	*Project use case*
Revision History:	*None*		
Objective:	*Verify that users can check the progress of a project.*		
Product/Ver/ Module:	*Project Management Module*		
Environment:	*Web-based application with stable internet connection.*		
Assumptions:	*User has appropriate permissions to view project progress.*		
Pre-Requisite:	*Logged into the application and project exists.*		
Step No.	**Execution description**	**Procedure result**	
	Navigate to the 'Projects' section.	*'Projects' page is accessible.*	
	Select the specific project to check progress.	*Project details page is displayed.*	
	Review the progress indicators..	*Progress of the project is displayed.*	
Comments: The test case is passed. Our system is working according to our document requirement.			
Passed			

6.2.2.6 Search Client

Table 6.19: Search Client Test Case

Search Client			
Test Case ID:	*6*	**QA Test Engineer:**	*Ali Hamza*
Test case Version:	*1*	**Reviewed By:**	*Hassan Khalid*
Test Date:	*25/4/2024*	**Use Case Reference(s):**	*Client use case*
Revision History:	*None*		
Objective:	*Verify that users can search for clients by specific criteria.*		
Product/Ver/ Module:	*Client Management Module*		
Environment:	*Web-based application with stable internet connection.*		
Assumptions:	*User has appropriate permissions to search clients.*		
Pre-Requisite:	*Logged into the application.*		
Step No.	**Execution description**	**Procedure result**	
	Navigate to the 'Clients' section.	*'Clients' page is accessible.*	
	Enter search criteria.	*Search criteria entered.*	
	Initiate the search.	*Search results are displayed.*	
Comments: The test case is passed. Our system is working according to our document requirement.			
Passed			

6.2.2.7 Logout

Table 6.20: Logout Test Case

Logout			
Test Case ID:	*7*	**QA Test Engineer:**	*Ali Hamza*
Test case Version:	*1*	**Reviewed By:**	*Hassan Khalid*
Test Date:	*25/4/2024*	**Use Case Reference(s):**	*Logout use case*
Revision History:	*None*		
Objective:	*Verify that users can successfully logout from the system.*		
Product/Ver/ Module:	*Authentication Module*		
Environment:	*Web-based application with stable internet connection.*		
Assumptions:	*User is currently logged in.*		
Pre-Requisite:	*User is currently logged in.*		
Step No.	**Execution description**	**Procedure result**	
	Locate the 'Logout' button or link.	*'Logout' option is accessible.*	
	Click on the 'Logout' button or link.	*User is logged out from the system.*	
	The user is redirected to the login page.	*User is redirected to the login page.*	
Comments: The test case is passed. Our system is working according to our document requirement.			
Passed			

6.3 Test Metrics

Table 6.21: Functionality Metrics

Functionality	**Check if all core functionalities of the system are working as intended.**
Number of Test Cases	**20**
Number of Test Cases Passed	**20**
Number of Test Cases Failed	**0**
Test Cases effectiveness	**20**

Chapter 7 User Manual

7.1 Client

7.1.1 Getting Started

To begin using Build Hub as a Client, follow these steps:

- Navigate to the Build Hub website's homepage.
- Log in using your email address and password on the login page.

7.1.2 Create New Project

To create a new Project on Build Hub, follow these steps:

- Log in to your Build Hub client account.
- Go to the "Add New Project" page.
- Fill out all the required fields carefully.
- Submit the form to create the new project listing.

7.1.3 Check Project Status

To check the status of a project, follow these steps:

- Log in to your Build Hub client account.
- Navigate to the "My Projects" page.
- Click on the specific project to see its status.

7.1.4 Chat with Others

To chat with others, follow these steps:

- Log in to your Build Hub client account.
- Navigate to the "Messages" page.
- Click on the specific person you want to chat with.

7.1.5 Bidding

To Bid on Projects, follow these steps:

- Log in to your Build Hub client account.
- Navigate to the “Bidding” page.
- Click on the specific project you want to bid on.

7.1.6 Use ChatBot

To use ChatBot, follow these steps:

- Log in to your Build Hub client account.
- Navigate to the “ChatBot” page.
- Chat with the AI.

7.1.7 Give Ratings to Builders

To give ratings to builders, follow these steps:

- Log in to your Build Hub client account.
- Navigate to the builder’s profile.
- Give ratings of your choice.

7.2 For Builder

7.2.1 Getting Started

To get started as a builder on Build Hub, follow these steps:

- Visit the Build Hub website’s homepage.
- Register for an account by providing all the required details.
- Once registered, you’ll be directed to your profile.

7.2.2 Give Response to Bids

To give a response to bids, follow these steps:

- Log in to your Build Hub builder account.
- Navigate to the “Bidding” page.
- Give a response to bids of your choice.

Chapter 8 Conclusion and work Completed

Significant turning points in the development of the construction industry have been reached by the Build Hub project. The platform, which allows for easy communication between manufacturers and clients, has modules for project matching, proposal submission, project costing, progress tracking, and cost estimation. Along with other hardware and software requirements, project important features, and system software requirements are provided. System er-diagrams and low-level and high-level setups are included in this report.

Build Hub projects can be made more functional by using other tools to estimate costs more accurately or by integrating sophisticated AI algorithms for more precise project matching. The user experience can also be improved by investigating augmented reality elements to display project progress and by increasing support for mobile applications. Two of the most important elements of our future growth strategy are adaptation to industry requirements and continuous improvement. It will be crucial to collaborate with industry leaders, incorporate cutting-edge technologies, and improve the user experience in order to guarantee that the Build Hub platform stays at the forefront of innovation in the building sector.

We have finished the frontend using a RESTful API, and we have constructed the full backend using Dotnet during Fyp 2. We have also concentrated on incorporating Firebase connection into our project. We have also implemented a recommendation system in Fyp 2. The implementation of AI Driven ChatBot will also be completed in 1st week of May.

Bibliography

[1] A. Smith and B. Johnson, "Digital platforms in construction management," *Journal of Construction Technology*, vol. 5, pp. 789–795, August 2023.

[2] L. Anderson, "Construction tech: The digital transformation," *Journal of Construction Innovation*, vol. 8, pp. 231–236, March 2023.

[3] R. Adams and S. Patel, "Machine learning in construction planning," *International Journal of Construction Engineering*, vol. 12, pp. 145–152, June 2023.

[4] D. Murphy, "Ocr for document verification in real estate," *Real Estate Technology Journal*, vol. 15, pp. 567–573, November 2023.

[5] E. White, "Internet's role in construction management," *Journal of Construction Management*, vol. 7, pp. 320–326, February 2023.

[6] F. Thompson, "Smart recommendations in real estate platforms," *Real Estate Technology Journal*, vol. 14, pp. 789–795, July 2023.

[7] G. Roberts, "User experience in digital construction platforms," *Journal of Construction Innovation*, vol. 9, pp. 411–418, September 2023.

[8] H. Lewis, "Cybersecurity in construction management platforms," *International Journal of Construction Engineering*, vol. 11, pp. 123–130, December 2023.

[9] I. Turner and J. Williams, "Digital transformation in real estate: A case study," *Real Estate Technology Journal*, vol. 13, pp. 234–241, April 2023.

[10] K. Martinez, "The future of construction: A digital forecast," *Journal of Construction Technology*, vol. 6, pp. 654–661, October 2023.